

Planning Access Point Placements

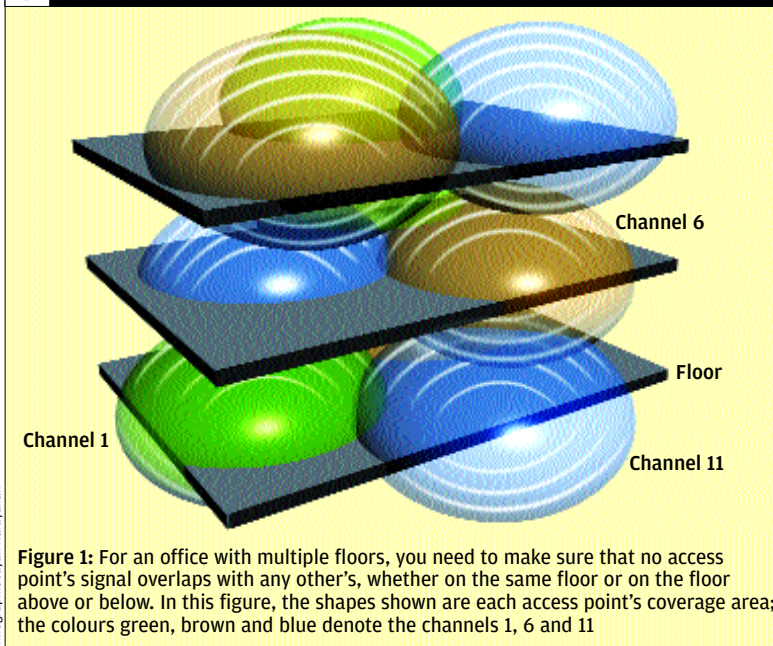


Figure 1: For an office with multiple floors, you need to make sure that no access point's signal overlaps with any other's, whether on the same floor or on the floor above or below. In this figure, the shapes shown are each access point's coverage area; the colours green, brown and blue denote the channels 1, 6 and 11

ference, the three standard channels used are 1, 6 and 11. To offer seamless roaming around the coverage area, you will need to ensure that any three neighbouring access points are configured to run on these three different channels.

This is easy when setting up a two-dimensional coverage area, say, on a single floor. However, when you have to do this for a building, with access required on multiple floors, sketch a 3D view of your coverage area and make sure that all access points—on all the floors—are set to operate on different channels. A good blueprint of your office building will come in handy. You will also need a wireless network monitoring tool to test the actual setup.

Bandwidth

Every network needs to provide its users with the bandwidth they require, and this is where wired LANs definitely have an edge. A standard wired LAN will offer your users a theoretical bandwidth of 100 Mbps, almost double the theoretical bandwidth of 54 Mbps for a WLAN. However, just as you do not expect the full 100 Mbps bandwidth from wired LANs, you cannot expect anywhere close to 54 Mbps for WLANs.

The most logical calculation is to halve this bandwidth and divide it by the number of users you plan to provide access to in a given area. For instance, if there are 10 users on a WLAN per access point, each user will get $27/10 = 2.7$ Mbps of maximum bandwidth, which will suffice for most. Only users who constantly transfer large files to and from a server will need more bandwidth. Take into account this user-density factor when deploying your WLAN.

Security

Now that you have your WLAN plans in place, your biggest concern will be security. No one wants to have their private office LAN visible to the outside world. Apart from people snooping around your LAN, you will also be put at risk by worms and viruses from neighbouring WLAN setups or wireless devices. What you need to do is ensure that your WLAN is as secure as it can be.

First of all, get a WLAN monitoring tool. In case your system administrators largely use Linux, then 'kismet', the default WLAN monitoring tool for Linux, will do the job. Windows users can choose from a huge list of commercial as well as free tools.

A seamless WLAN will need all access points to have the same Service Set Identifier (SSID). The primary thing systems administrators should do is change the SSID so as to leave others guessing. Default SSIDs for various

The Interference Dilemma

The biggest thing that afflicts a WLAN is interference. A WLAN signal is fragile and can easily be disrupted by something as simple as a wall, so the last thing you need are pesky devices sending your WLAN crashing. The 802.11 protocol runs at about 2.4 GHz, and this frequency is prone to interference from numerous sources. The problem with interference is the additional load that is put on your networks. If one wireless device is sending data packets to another, and interference occurs, the receiver gets a packet with errors.

Most often, receivers are programmed to not respond to the sender when this happens, and this results in the sender transmitting the packet once again. The problem here is that all these packets being re-sent can use as much as half your bandwidth, which will hurt the overall WLAN performance. Some devices that can cause interference with WLANs are:

Cordless phones running at 2.4 GHZ

These are the latest cordless phones that offer a longer range from their base station. However, the problem lies in the fact that these cordless phones use the same frequency as WLANs, and thus turn out to be the biggest source for interference. If you have any of these phones installed in your office, you will need to get rid of them whenever you install your WLAN or be ready to face pathetic speeds.

Bluetooth-enabled devices

Many mobile phones and PDAs today are Bluetooth-enabled, and this can cause interference if a user uses such a device where the signal strength of an access point is already low—far away from the access point and close to a device connected to the WLAN. Thankfully, current research, driven by the increase in popularity of both Bluetooth-enabled

devices and Wi-Fi, will soon put an end to this menace.

Microwave ovens

Every time employees heat their food, you might find users in the immediate surroundings yanking out clumps of hair. Placing an access point next to a microwave oven is a definite no-no, as these ovens emit signals that clash with Wi-Fi.

Power Cables

Improperly shielded power cables can interfere with wireless signals, and it is best to keep access points positioned away from power cables or power junction boxes.

Other WLANs

Most companies in India share buildings, floors or even offices with other companies. In such a scenario, a neighbouring office's WLAN could cause interference with your own deployment. This is something you should inspect and test thoroughly when doing your RF site survey.